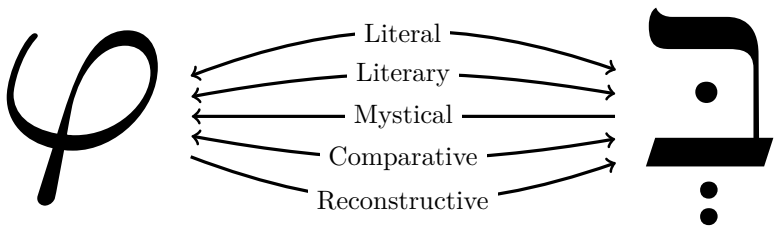
How should physical cosmology shape our interpretation of the creation narrative?

Outline:

1. Five frameworks for discussion
2. Physical cosmology on one foot
3. Scientific insights
4. Takeaways: גם זה תורה



Interpretive frameworks:

A. *Literal:* Should the narrative correspond with scientific chronology? What is the alternative meaning?
B. *Literary:* How do differences reflect changes in our language, labels, and modes of expression for creation?
C. *Mystical:* What does Kabbalistic wisdom reveal about the meaning of scientific findings?
D. *Comparative:* How does the scientific story add to the dialectic driven by existing tensions in the narrative?
E. *Reconstructive:* From the scientific perspective, how would *we* write the story of creation for this audience?

Genesis בראשית

1:1 In the beginning God created the heaven and the earth. And the earth was without form and void; and darkness was on the face of the deep. And a wind from God moved over the surface of the waters. And God said, Let there be light: and there was light. And God saw the light, that it was good: and God divided the light from the darkness.

א:א בְּרֵאשִׁית בָּרָא אֱלֹקִים אֵת הַשָּׁמַיִם וְאֵת הָאָרֶץ: וְהָאָרֶץ הָיְתָה תֹהוּ וָבֹהוּ וְחֹשֶׁךְ עַל־פְּנֵי תְהוֹם וְרוּחַ אֱלֹקִים מְרַחֶפֶת עַל־פְּנֵי הַמָּיִם: וַיֹּאמֶר אֱלֹקִים יְהִי אוֹר וַיְהִי־אוֹר: וַיַּרְא אֱלֹקִים אֶת־הָאוֹר כִּי־טוֹב וַיַּבְדֵּל אֱלֹקִים בֵּין הָאוֹר וּבֵין הַחֹשֶׁךְ:

1

Choose your own adventure! Pick one or more of the frameworks above to connect the text with the scientific description of cosmology on page 4. The sources below provide relevant arguments and examples.

Maimonides, *Guide for the Perplexed* רמב״ם, מורה הנבוכים

2:29:27 First, the account given in Scripture of the Creation is not, as is generally believed, intended to be in all its parts literal. […] The literal meaning of the words might lead us to conceive corrupt ideas and to form false opinions about God, or even entirely to abandon and reject the principles of our Faith. It is therefore right to abstain and refrain from examining this subject superficially and unscientifically.

ב:כט:כז אחת מהם – זאת ההקדמה, והיא שכל מה שנזכר ב׳מעשה בראשית׳ ב׳תורה׳ אינו כולו על פשוטו כפי מה שידמו ממנו ההמון […] כי הפשוטים ההם מביאים אם להפסד דמיון גדול והרכבת דעות רעות בחוק האלוה או לביטול גמור וכפירה ביסודי התורה. והנכון – לעזוב ולהרחיק בחינתם בדמיון לבד והנטיה מן החכמות, ולא כמו שיעשו ה׳דרשנין׳

2

3:14:6 You must, however, not expect that everything our Sages say respecting astronomical matters should agree with observation, for mathematics were not fully developed in those days: and their statements were not based on the authority of the Prophets, but on the knowledge which they either themselves possessed or derived from contemporary men of science.

ג:יד:ו ולא תבקש ממני שיסכים כל מה שזכרוהו מענין התכונה למה שהענין נמצא – כי החכמות הלימודיות היו בזמנים ההם חסרות, ולא דברו בהם על דרך קבלה מן הנביאים, אבל מאשר הם חכמי הדורות ההם בענינים ההם, או מאשר שמעום מחכמי הדורות ההם.

3

Bereishit Rabbah בראשית רבה

8:1 Rabbi Shimon ben Lakish said: [The creation of Adam was] last among the acts of creation on the last day, and first among the acts of creation on the first day. […] If a person is meritorious, it is said to him: 'You preceded the ministering angels'; if not, it is said to him: 'A fly preceded you, a gnat preceded you, this earthworm preceded you.'

ח:א אָמַר רַבִּי שִׁמְעוֹן בֶּן לָקִישׁ, אָחוֹר לְמַעֲשֶׂה יוֹם הָאַחֲרוֹן, וָקֶדֶם לְמַעֲשֶׂה יוֹם הָרִאשׁוֹן […] אִם זָכָה אָדָם אוֹמְרִים לוֹ אַתָּה קָדַמְתָּ לְמַלְאֲכֵי הַשָּׁרֵת, וְאִם לָאו אוֹמְרִים לוֹ זְבוּב קְדָמְךָ, יַתּוּשׁ קְדָמְךָ, שְׁלִשׁוּל זֶה קְדָמְךָ.

4

Genesis בראשית

1:11 And God said, Let the earth bring forth grass, herb yielding seed, and fruit tree yielding fruit after its kind, whose seed is in itself, upon the earth: and it was so. […] Let there be lights in the firmament of heaven to divide the day from the night […]

א:יא וַיֹּאמֶר אֱלֹקִים תַּדְשֵׁא הָאָרֶץ דֶּשֶׁא עֵשֶׂב מַזְרִיעַ זֶרַע עֵץ פְּרִי עֹשֶׂה פְּרִי לְמִינוֹ אֲשֶׁר זַרְעוֹ־בוֹ עַל־הָאָרֶץ וַיְהִי־כֵן: […] וַיֹּאמֶר אֱלֹקִים יְהִי מְאֹרֹת בִּרְקִיעַ הַשָּׁמַיִם לְהַבְדִּיל בֵּין הַיּוֹם וּבֵין הַלָּיְלָה […]

5

Nachmanides on Genesis רמב״ן על בראשית

1:11:1 He decreed that there be among the products of the earth a force which grows and bears seed so that the species should exist forever.

א:יא:א גזר שיהיה בתולדות הארץ כח הצומח ומוליד זרע כדי שיהיה המין קיים לעד

6

Genesis בראשית

1:27 So God created Mankind in His own image, in the image of God He created him; male and female He created them.

א:כז וַיִּבְרָא אֱלֹקִים | אֶת־הָאָדָם בְּצַלְמוֹ בְּצֶלֶם אֱלֹקִים בָּרָא אֹתוֹ זָכָר וּנְקֵבָה בָּרָא אֹתָם:

7

2:7 And the Lord God formed man of the dust of the ground, and breathed into his nostrils the breath of life; and man became a living soul.

ב:ז וַיִּיצֶר ה׳ אֱלֹקִים אֶת־הָאָדָם עָפָר מִן־הָאֲדָמָה וַיִּפַּח בְּאַפָּיו נִשְׁמַת חַיִּים וַיְהִי הָאָדָם לְנֶפֶשׁ חַיָּה:

8

2:21 And the Lord God caused a deep sleep to fall upon the man, and he slept: and He took one of his sides, and closed up the flesh in its place, and of the side, which the Lord God had taken from the man, He made a woman, and brought her to the man.

ב:כא וַיַּפֶּל ה׳ אֱלֹקִים | תַּרְדֵּמָה עַל־הָאָדָם וַיִּישָׁן וַיִּקַּח אַחַת מִצַּלְעֹתָיו וַיִּסְגֹּר בָּשָׂר תַּחְתֶּנָה: וַיִּבֶן ה׳ אֱלֹקִים | אֶת־הַצֵּלָע אֲשֶׁר־לָקַח מִן־הָאָדָם לְאִשָּׁה וַיְבִאֶהָ אֶל־הָאָדָם:

9

Bereishit Rabbah בראשית רבה

14:3 There were two acts of formation [involved in the creation of man], a formation from the lower realms and a formation from the upper realms.

יד:ג שְׁתֵּי יְצִירוֹת, יְצִירָה מִן הַתַּחְתּוֹנִים וְיִצִירָה מִן הָעֶלְיוֹנִים.

10

14:4 [The two acts of formation (yetzirah), allude to the creation of man's] two inclinations (yetzarim), the good inclination and the evil inclination.

יד:ד שְׁנֵי יְצָרִים, יֵצֶר טוֹב וְיֵצֶר הָרָע.

11

Maimonides, *Guide for the Perplexed* רמב״ם, מורה הנבוכים

2:30:7 The four elements indicated, according to our explanation, in the term "earth," in the first verse, are mentioned first after the heavens: for there are named ereẓ (earth), ruaḥ (air), mayim (water), and hoshek (fire).

ב:ל:ז וממה שצריך שתדעהו – כי היסודות הארבע אשר אמרנו ששם ׳ארץ׳ הראשון יורה עליהם נזכרו תחלה אחר השמים. שהוא זכר ׳ארץ׳ הראשון יורה עליהם נזכרו תחילה אחר השמים. שהוא זכר ׳ארץ׳ ו׳מים׳ ו׳רוח׳ ו׳חושך׳

12

Nachmanides on Genesis רמב״ן על בראשית

1:1:3 He brought forth from total and absolute nothing a very thin substance devoid of corporeality but having a power of potency, fit to assume form and to proceed from potentiality into reality. This was the primary matter created by G-d; it is called by the Greeks *hyle* (ὕλη). After the hyle, He did not create anything, but He formed and made things with it […] This substance, which the Greeks called hyle, is called in the sacred language *tohu*…

א:א:ג הוציא מן האפס הגמור המוחלט יסוד דק מאד, אין בו ממש, אבל הוא כוח ממציא, מוכן לקבל הצורה ולצאת מן הכוח אל הפועל. והוא החומר הראשון, נקרא ליוונים "היולי". ואחר ההיולי לא ברא דבר, אבל יצר ועשה […] והחומר הזה, שקראו היולי, נקרא בלשון הקדש "תהו" […]

13

Aristotle, *Metaphysics*, 7.1032a

Ἀριστοτέλης, *τὰ μετὰ τὰ φυσικά*, Z΄ΑΛΒ

Natural generation is the generation of things whose generation is by nature. That from which they are generated is what we call matter (ὕλην); that by which, is something which exists naturally; and that which they become is a man or a plant or something else of this kind, which we call substance in the highest degree. All things which are generated naturally or artificially have matter; for it is possible for each one of them both to be and not to be, and this possibility is the matter in each individual thing. (14)

Judah Halevi, The *Kuzari*

יהודה בן שמואל הלוי, ספר הכוזרי

1:64 *Al Khazari:* Does this mean that Aristotle's philosophy is not deserving of credence? *The Rabbi:* Certainly. He exerted his mind, because he had no tradition from any reliable source at his disposal. […] Had he lived among a people with well authenticated and generally acknowledged traditions, he would have applied his deductions and arguments to establish the theory of creation, however difficult, instead of eternity, which is even much more difficult to accept.

א:סד אָמַר הַכּוּזָרִי: וְכִי זֶה מְחַיֵּב שֶׁלֹּא נַאֲמִין לַאֲרִיסְטוּ בְּחָכְמָתוֹ. אָמַר הֶחָבֵר: בֵּן, מִפְּנֵי שֶׁהִטְרִיחַ אֶת שִׂכְלוֹ וּמַחֲשַׁבְתּוֹ בַּעֲבוּר שֶׁלֹּא הָיְתָה בְּיָדוֹ קַבָּלָה מִמִּי שֶׁיַּאֲמִין בְּהַגָּדָתוֹ […] וְאִלּוּ הָיָה הַפִּילוֹסוֹף בְּאֻמָּה שֶׁיֵּנָחֵל מְקֻבָּלוֹת וּמְפֹרָסָמוֹת שֶׁאֵינוֹ יָכוֹל לִדְחוֹת אוֹתָם, הָיָה מִתְעַסֵּק בְּהַקָּשׁוֹתָיו וּמוֹפְתָיו לְהַחֲזִיק הַחִדּוּשׁ עִם קָשְׁיוֹ. כַּאֲשֶׁר הֶחֱזִיק הַקַּדְמוּת אֲשֶׁר הוּא יוֹתֵר קָשֶׁה לְקַבֵּל. (15)

Abraham Isaac Kook, *Igrot HaRe'aya* 91

אַבְרָהָם יִצְחָק הַכֹּהֵן קוּק, אגרות הראי״ה צ״א

The Torah related the story of creation in a closed manner and spoke in hints and parables. We know that the actual story behind creation is one of the Torah's deepest secrets (see Chagiga 11b). […] The main thing is the "knowledge of Hashem" that comes out of the episode and learning true morality. […] Similarly that which is uncovered over thousands of years about the manner of the world's development, which increases our knowledge, is still revealed to us cryptically. The main point is that there is a time and place for each matter, without Hashem leaving anything to chance. (16)

Samson Raphael Hirsch, *Translation and explanation of the Pentateuch*

[…] Creation was from nothing, a truth that forms the foundation of the consciousness that God's teachings seek to build for us. The opposite, the eternity of the material of the world, which only allows the Creator to be the creative designer and which also forms the foundation of pagan consciousness to this day, is not only a metaphysical lie that robs people's cosmogonic ideas of truth, i.e., their agreement with reality: it is the even more pernicious denial of all freedom in God and in man, which undermines all morality. If the world creator had been given the material, he could not create the absolutely good world from the given material, but only the relatively best world. (17)

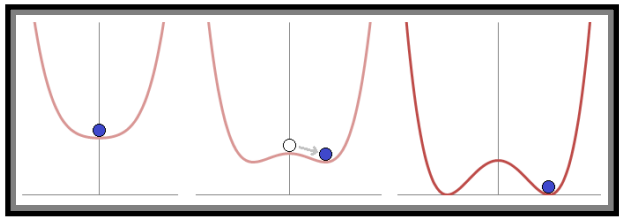
Joseph Soloveitchik, *The Lonely Man of Faith*

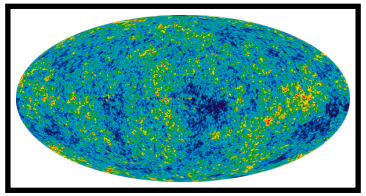
I Let us portray these two men, Adam the first and Adam the second, in typological categories. […] Adam the first is interested in just a single aspect of reality and asks one question only — "How does the cosmos function at all?" nor is he interested in the question, "What is its essence?" He is only curious to know how it works. […] Adam the first wants to be human, to discover his identity which is bound up with his humanity. […] Adam the first is the mathematical scientist who whisks us away from the array of tangible things […] into a formal relational world of thought constructs… (18)

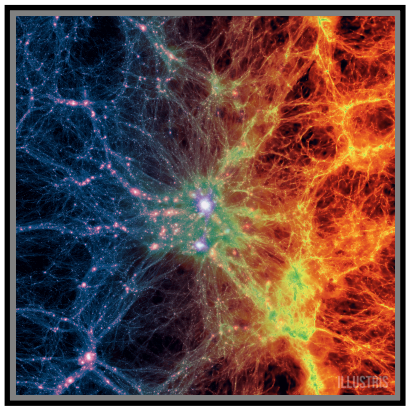
III Adam the second is, like Adam the first, also intrigued by the cosmos. […] His inquiry is of a metaphysical nature and a threefold one. He wants to know: "Why is it?" "What is it?" "Who is it?" […] He wants to understand the living, "given" world into which he has been cast. Therefore, he does not mathematize phenomena or conceptualize things. He encounters the universe in all its colorfulness, splendor, and grandeur… (19)

IX The Biblical dialectic stems from the fact that Adam the first, majestic man of dominion and success, and Adam the second, the lonely man of faith, obedience, and defeat, are not two different people locked in an external confrontation as an "I" opposite a "thou", but one person who is involved in self-confrontation. […] Since the dialectical role has been assigned to man by God, it is God who wants the man of faith to oscillate between the faith community and the community of majesty, between being confronted by God in the cosmos and the intimate, immediate apprehension of God through the covenant, and who therefore willed that complete human redemption be unattainable. (20)

Physical cosmology on one page — קוסמולוגיה פיזית על נייר אחד









Electroweak symmetry breaking[4]
ויהי אור

Recombination[7]
ויבדל בין האור ובין החשך
$t \sim 300{,}000$ years

Present day
$t \approx 13.7$ billion years

singularity[1] (??)
ראשית

Nucleosynthesis[6]
$t \sim 1$ second

Pop. III stars[10]
יהי מאורות ברקיע

"Cosmic noon"[12]
$t \sim 3$ billion years

time

Inflation[2]

Reheating[3]
תהו ובהו

Hadronization[5]
החומר הראשון / ὕλην

Dark ages[8]

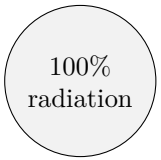
Structure formation[9]
יהי רקיע בתוך המים

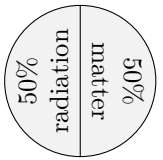
Reionization[11]
$t \sim 1$ billion years

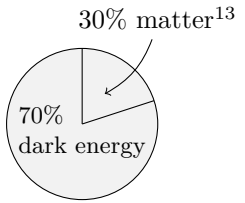
Earth forms
$t \sim 10$ billion years











[1] A singularity means a pointlike Universe. Our theories break down, so we don't know what happened here.
[2] A hypothesized period of exponential expansion. This would be driven by an *inflaton* field.
[3] The end of inflation, when energy would be dumped into a hot soup of particles. This is the "Big Bang."
[4] A phase transition in the fundamental fields of the universe not unlike water freezing. Photons start here.
[5] The Universe cools enough that quarks can no longer exist on their own, and form hadrons (e.g. protons).
[6] Protons and neutrons begin forming light atomic nuclei (H, He, Li, Be). These are ions at this point.
[7] Electrons bind to nuclei, making atoms. Light can now propagate freely as cosmic background radiation.
[8] During this epoch, there are no sources of light apart from cosmic background radiation.
[9] Collisionless dark matter clouds collapse and pull in gas, forming big clumps of matter amid the void.
[10] The gas cools and collapses, making the first stars ("Population III") out of just hydrogen and helium.
[11] Light from stars and accreting black holes ionizes the gas in the Universe.
[12] The formation rate of stars and galaxies peaks. Stars begin producing "heavy" elements (like carbon).
[13] Of this 30% matter component, 80% is invisible, collisionless dark matter.

Motivating questions:

1. Does a detailed picture of cosmology introduce new tensions at the level of the פשט ?
2. Even in metaphor, the textual narrative proceeds from chaos to order. What does that mean in cosmology?
3. How does the notion and origin of substance compare between בראשית and cosmic chronology?
4. What problems in the philosophy of cosmology does the text *not* anticipate? Does it supply answers?